

115.109

Introduction to Business Analytics

Welcome!

Dr Andrew Murphy
(Please contact Andrew via
Stream's Communication Tools)

Dr Andrew Murphy
Instructor



Pepeha (Personal Introduction)



Ko Puketāpapa te maunga
Ko Pupuke te roto
Ko Waitematā te moana
Ko Andrew Murphy tōku ingoa

He mihi Rangatira tenei
Kia koutou katoa
No reira

Tēnā Koutou, Tēnā Koutou, Tēnā Koutou Katoa



Your Lecturers:

- Course Coordinator:
- Dr. Andrew Murphy



- Weekly guest lecturers, e.g:
Kitty Ling (Senior Manager Analytics Transformation, Woolworths NZ)



Andrew Arnold
Shopper Intelligence

Aim: **NOT (JUST) A STATS PAPER!**

“An introduction to internal and external organisational data, mechanisms to obtain, analyse and interpret that data, and performance indicators used to define and assess the operational and financial viability of a business”

Students who successfully complete this paper should be able to:

- ***Describe how external information can be used to assess a business or market.***
- ***Outline how data can be converted into meaningful information, and patterns and opportunities recognised within that data.***
- ***Identify how information can be interpreted to inform and improve business decision-making.***

Definitions:

an introduction to business analytics

An introduction: no prior depth knowledge expected.

Familiarity with maths and statistics helpful, but not essential.

Familiarity with Excel very helpful, but not essential.

to business: the focus in this paper will be largely but not only on retail businesses – ideas and metrics apply to many contexts

(data): take a wide variety of forms and sources. We will primarily focus on structured numerical data, but also consider unstructured and qualitative data

analytics: this is not a statistical theory class, but we will use some statistical concepts and tests – and other analytical tools – using Excel

NOT ADVANCED ANALYTICS:

➤ Welcome to Advanced Customer Analytics

Advanced Customer Analytics is a unique marketing research consultancy. We integrate advanced statistical analysis with an in-depth understanding of purchase motivations and marketing techniques. Within the marketing and survey research realm, we provide research design consulting and advanced statistical analysis, focusing primarily on the customer experience, customer segmentation and brand health. <http://advancedcustomeranalytics.com>

- Structural equation models
- Partial least squares
- Logistic regression
- Machine learning
- Discrete choice & Conjoint
- TURF (Total Unduplicated Reach and Frequency)
- Cluster analysis
- Latent class regression
- CHAID (CHi-squared Automatic Interaction Detection)
- Time series analysis and forecasting
- Survival analysis

*See **Business Analytics Major**
Including:
157216 – Management, Analytics
and Decision making
115326 Data Visualisation*

Expectations

- **Attend (view) lectures on time having prepared the reading specified in the course guide;**
- **Read widely about data and data analytics, including the media and academic articles;**
- **Complete and submit computer labs regularly and before due date, helping others if requested (eg in the section forum);**
- **Note: there is often a very high correlation between attendance/participation and good grades, so you are strongly advised to attend/participate regularly.**
- **This is NOT a course you can leave until the end and hope to catch up.**

Methods

- ~ 2 hrs live (& recorded) lectures per week over 12 weeks
plus: High profile guest speaker recordings
- In-person computer lab support session (Albany): Wednesdays 4-5pm
- Live computer lab support session via Zoom Wednesdays 8-9pm
Zoom link: ID: 83374275118; passcode 115109 (see online classroom)
- Self study through reading (textbook and supplementary readings)
- Assessment of understanding via coursework

Stream

Assessment	Learning Outcomes	Percentage Weighting
A1: Computer Laboratories (5 file submissions)	1,2,3	(10% each) 50%
A2: Test 1 (Topics 1-5)	1,2	10%
A3: Test 2 (Topics 6-10)	1,2	10%
A4: Written report (group)	1,2,3,4	20%
A4: Individual reflection	1,2,3,4	10%

Assessment Planner

Assessment	Due Dates
Computer labs weeks 1,2 (A1) including Red Box Quizzes	August 03
Computer labs weeks 3,4 (A1) including Red Box Quizzes	August 17
Test 1 (A2)	Friday 14 Aug to Friday 21 Aug (W6)
Computer labs weeks 5,6 (A1) including Red Box Quizzes	August 31
Computer labs weeks 7,8 (A1) including Red Box Quizzes	October 05
Test 2(A3)	Friday 2 Oct to Friday 9 Oct (W11)
Computer labs weeks 9,11 (A1) including Red Box Quizzes	October 19
Group Report (A4) and individual reflection (A4)	Monday 19 October
Fortnightly quiz assessment (optional Formative Assessment)	Every two weeks through the semester

Course Guide

Assessment	Learning Outcomes	Percentage Weighting	Due Dates
Computer labs weeks 1,2 including Red Box Quiz 1,2 (A1)	1, 2, 3	8+2%	August 03
Computer labs weeks 3,4 including Red Box Quiz 3,4 (A1)	1, 2, 3	8+2%	August 17
Test 1 (Assessment 2)	1, 2	10%	Friday 14 August to Friday 21 August (W6)
Computer labs weeks 5-6 including Red Box Quiz 5,6 (A1)	1, 2, 3	8+2%	August 31 (during the mid-semester study period)
Computer labs weeks 7,8 including Red Box Quiz 7,8 (A1)	1, 2, 3	8+2%	October 05
Test 2 (A3)	1, 2	10%	Friday 02 October to Friday 09 October (W11)
Computer labs weeks 9,11 including Red Box Quiz 9,11 (A1)	1, 2, 3	8+2%	October 19
Project Report (A4)	1, 2, 3, 4	30%	October 19 (W12)
Fortnightly quiz assessment (Formative Assessment)	1, 2, 3	0%	Every two weeks through the semester

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assessments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Green and Red Boxes

- To help you determine if you are on the right track when completing the lab tasks, some cells in the Excel worksheets are highlighted in yellow.
- When you enter a correct value (or close enough, if rounding could affect the workings), the cell should turn green. If it does not, check your workings. Ask Andrew (in case there is an error in what the conditional formatting is set to).
- These Green Boxes count for 'overall impression' marks for the lab.

This is what the Red Box Quiz looks like: the Task / step and cell reference are identified, and the number of expected decimal places to enter

	A	B	C	D	E	F	G	H	I	J
16										
17					(or close enough, if rounding could affect precision of answer). If it stays yellow, check your workings.					
18										
19					Answer	Sh				
20		1.1 What is the ex-GST price?	Hint: =439.99/1.15		382.60	=4				
21		1.2 What is the GST included in the RRP?				=E				
22										
23		If it costs \$200 (ex-GST) to add the bike to inventory	COGS:		200					
24		1.3 What is the Gross Margin? (GM)								
25		1.4 What is the GM%?								
26										
27		Recalculate 1.3-1.4 if the next purchase order raises the per unit acquisition cost (COGS								
28		1.5 What is the COGS for the latest order?				(N				
29		1.6 What is the Gross Margin \$?								
30		1.7 What is the Gross Margin %?								
31		1.8 What is the markup %?								
32										
33		Task 2:								

Question 1

Not yet answered Marked out of 1.00 Flag question

1.4: What is the Gross Margin % (GM%)? (cell E25) {round answer to 1 decimal place [1 d.p.], do not include the % sign}

Answer:

Do not use the % formatting, as the box will not turn green if you do, even if it shows the right answer

Hint: GM / COGS * 100

- Each Lab also has a Red Box Quiz. These count for 20% of the lab marks (so "Lab 1 Red Box Quiz" counts for 1% of the course total, or 1 mark), with the remaining 80% being assessed from the Lab file submission (so the Lab 1 sheet counts for 4% of the course).
- The Red Box Quizzes can only be attempted once, and you will only learn of the correct answers once the Lab file submission window has closed.

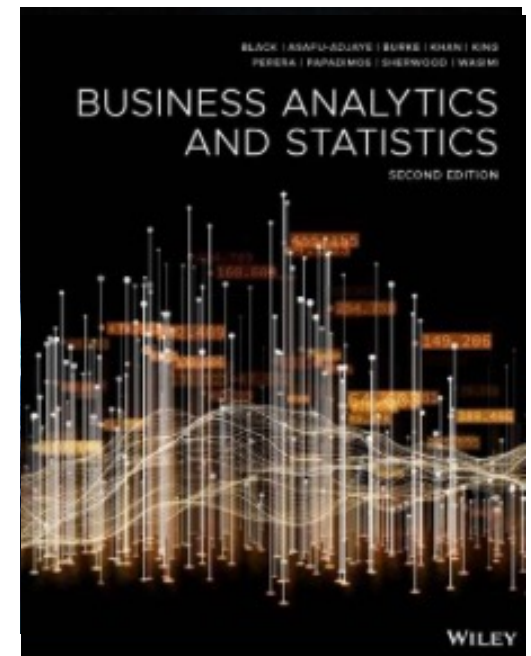
Readings

Highly recommended text:

Black K, Asafu-Adjaye J, Burke P, Perera N, Sherwood C & Wasimi S. (2018) Business Analytics And Statistics 1st ed (Wiley)

Black K, Asafu-Adjaye J, Burke P, Khan N, King G, Perera N, Papadimos A, Sherwood C & Wasimi S. (2024) Business Analytics And Statistics 2nd ed (Wiley)

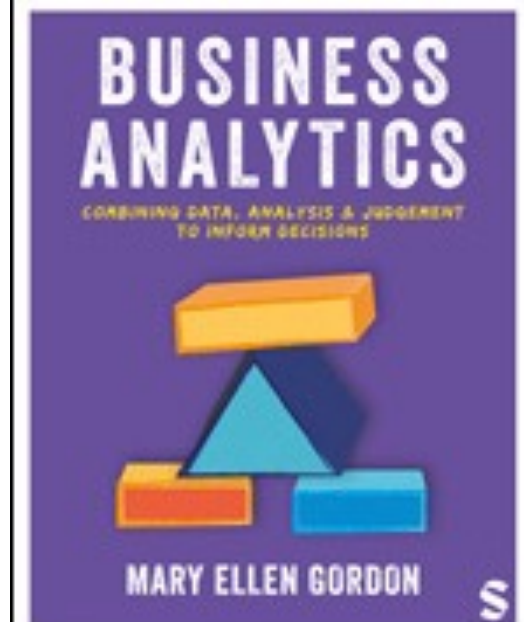
Special offer of A\$19.95 for Wiley Business Now for the semester



Also recommended:

Gordon M-E (2023): Business Analytics: combining data, analysis & judgement to inform decisions (Sage)

- Specific readings on particular topics will be put on Stream
- See Course Guide for a list of suitable academic journals to read from



Many other options:

BUSINESS ANALYTICS, 3RD EDITION Paperback – January 1, 2021

by James Evans (Author)

★★★★★ ∨ 11 ratings

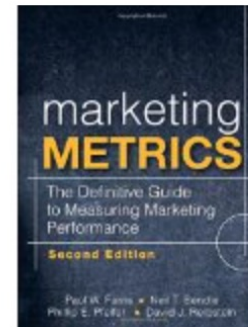
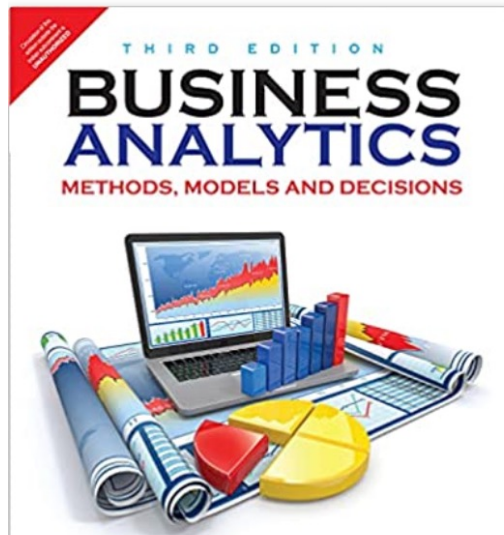
[See all formats and editions](#)

Paperback

\$50.32

6 Used from \$36.39

20 New from \$30.59



Marketing Metrics: The Definitive Guide to I

by Paul W. Farris and Neil T. Bendle

Hardcover

\$34.58 ~~\$54.99~~ ✓Prime

Get it by **Wednesday, May 6**

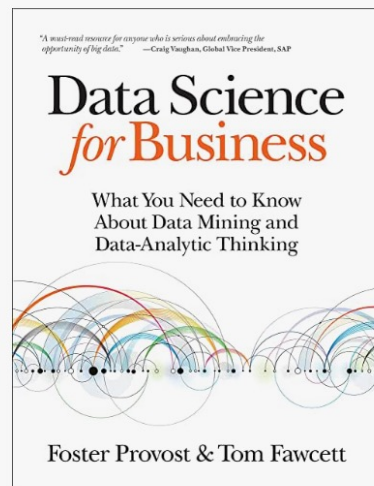
More Buying Choices

\$27.99 used & new (62 offers)

Kindle Edition

\$39.98

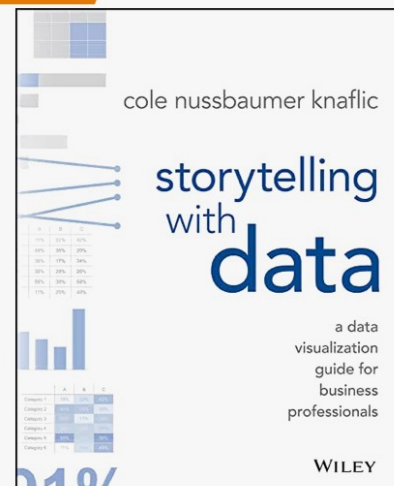
Auto-delivered wirelessly



Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking

by Foster Provost and Tom Fawcett

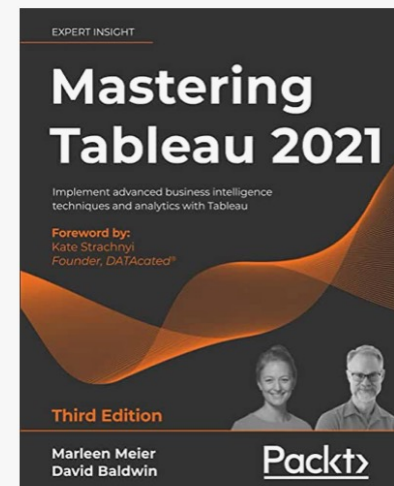
Best Seller



Storytelling with Data: A Data Visualization Guide for Business Professionals

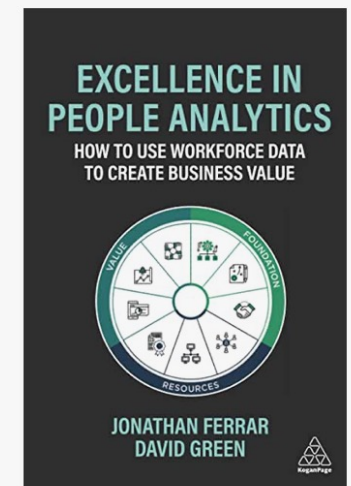
by Cole Nussbaumer Knaflic

★★★★★ ∨ 3.131



Mastering Tableau 2021: Implement advanced business intelligence techniques and analytics with Tableau, 3rd Edition

by Marleen Meier . David Baldwin. et al.



Excellence in People Analytics: How to Use Workforce Data to Create Business Value

by Jonathan Ferrar and David Green

★★★★★ ∨ 57

Assessment

- Coursework (100%)
 - Two tests of readings, lectures and practical work (2x10%): first test online from Friday 14/8 to 21/8 (W6), second test from Friday 2/10 to 9/10 (W11)
 - Group Project (30%) due Monday 19 October
 - 5x computer lab submissions each worth 10% (=50%) including 10 Red Box Quizzes
- See Assessment section in Course Guide and Stream for further information

Withdrawal dates

The last day to withdraw from this course:

- Without financial penalty: 3 August 2026
- Without academic penalty: 15 October 2026

Class Reps Needed

Sign Up to be a Class Representative

What will being a Class Representative require?

- Attendance at an introductory training session – you'll receive an invitation from your students' association
- Some initiative:
 - introduce yourself to your classmates and lecturer
 - be approachable
 - address issues or report them to your students' association in a proactive manner etc
- Completing online fortnightly logbook feedback for your courses
- You may also be offered additional optional advanced training modules

Signing Up

<https://portal.tetiraahupae.ac.nz>

Signing up is quick and easy to do!

Remember that there can be more than one Class Representative in a class, in fact, we recommend it!

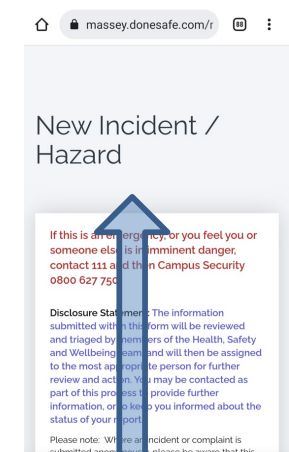
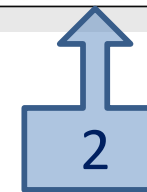
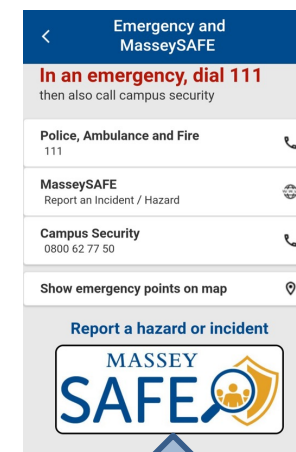
Evacuation upon hearing the alarm

- Leave room IMMEDIATELY by nearest safe exit taking only essential personal items with you.
- If safe to do so, help others who need assistance.
- Do not run and do not use lifts.
- Follow EXIT signs  and go to the evacuation assembly area which is located
- Follow instructions from Wardens (wearing bright yellow or orange vests).
- DO NOT re-enter building until an “all clear” is given.
- Inform the building warden if anyone is left in the building.



Injury / Near-miss Reporting

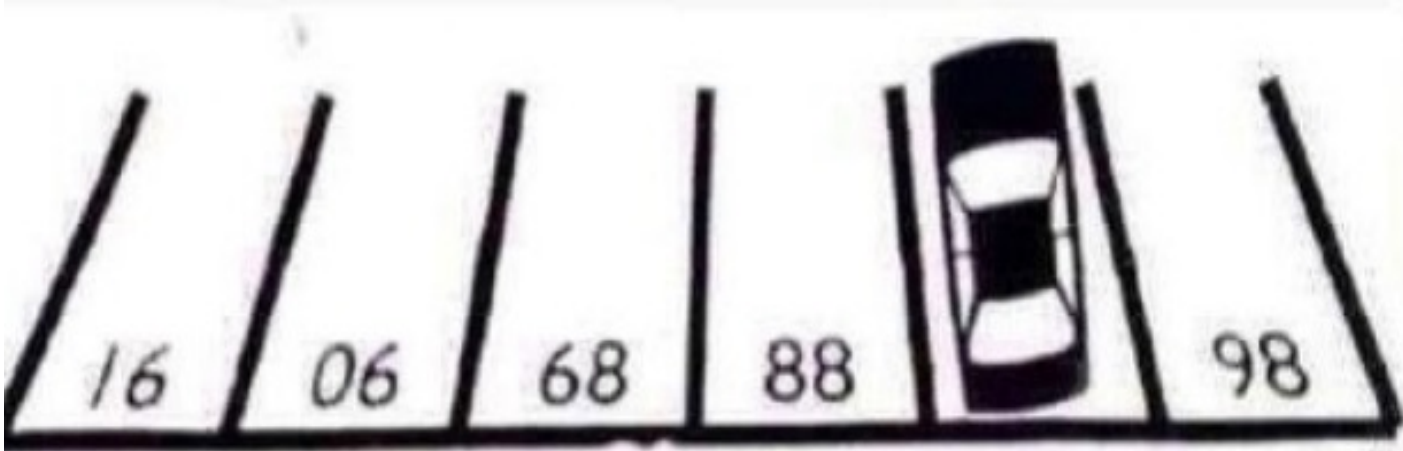
- If you injure yourself on campus, in class, at the gym, in the grounds, on field trips etc or have a near-miss - Report it using the Massey Uni app.
- Inform your manager, technician or academic supervisor.
- Seek first aid assistance, if needed.



Smoke Free and Vape Free Policy

- Campus buildings and grounds are smoke free and vape free, with the exception of designated shelters.
- All staff, students, visitors and contractors are expected to abide by the Smoke Free and Vape Free policy.
- Disciplinary action may occur if the policy is breached.

Puzzle

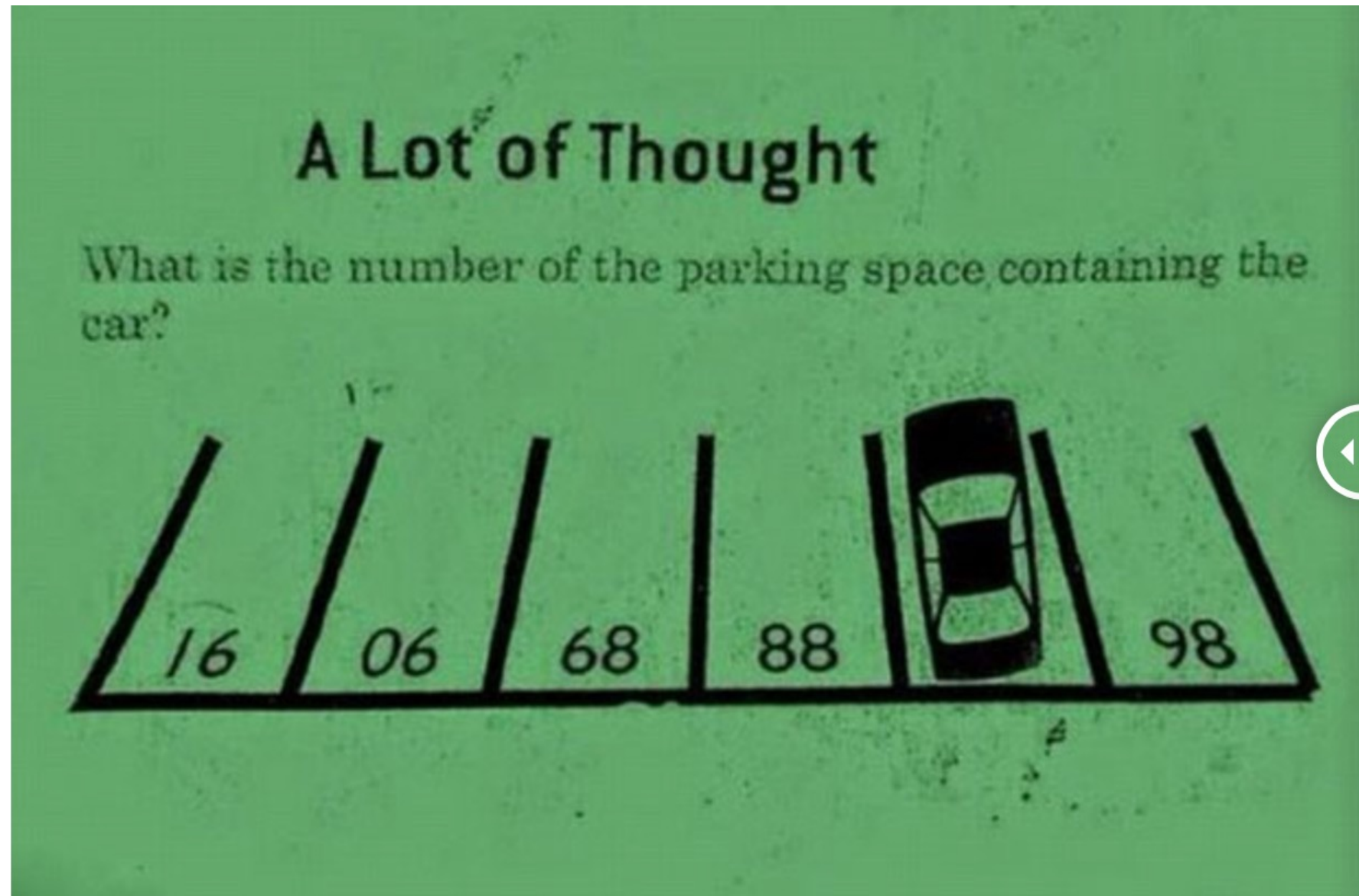


香港小学入学考试题: 21题 Hong Kong Elementary School Admissions Test Question: #21

What parking spot # is the car parked in?
请问汽车停的是几号车位?

请在20秒内完成回答 Please answer within 20 seconds. @微博搞笑排行榜

Puzzle



Puzzle

- **Problems can appear challenging, possibly unsolvable**
- **We can try lots of techniques to solve them: sequences, patterns, formulae, statistics...**
- **Sometimes the solution lies in looking at the problem from a different angle**

Puzzle

https://www.nzherald.co.nz/lifestyle/are-you-smarter-than-a-six-year-old-take-this-test/T7GVCHJYBCAZIJE6QGYC47GYFA/?c_id=6&objectid=11467301



Global Powers of Retailing Top 250, FY2023 (2025 report)

<https://www.deloitte.com/content/dam/assets-shared/docs/industries/consumer/2025/deloitte-global-powers-of-retailing-2025.pdf>

FY2023 retail revenue rank	Rank change from FY2022 ¹	Name of company	Country of origin	FY2023 retail revenue (US\$m)	FY2023 parent company/ group revenue ² (US\$m)	Dominant operational format	# countries of operation	FY2018- 2023 retail revenue CAGR ³	FY2022- 2023 retail revenue growth	FY2023 Net profit margin ⁴
1	● 0	Walmart Inc.	United States	648 125	648 125	Hypermarket/ supercenter	19	4.7%	6.0%	2.5%
2	● 0	Amazon.com, Inc.	United States	251 902	574 785	Non-store	23	12.4%	5.4%	5.3%
3	● 0	Costco Wholesale Corporation	United States	242 290	242 290	Cash & carry/ warehouse club	14	11.3%	6.8%	2.6%
4	● 0	Schwarz Group	Germany	177 009	181 015	Discount store	32	9.4%	8.9%	n/a
12	▲ 1	Ahold Delhaize	Netherlands	97 837 **	97 837 **	Supermarket	10	7.1%	1.9%	2.1%
13	▲ 1	Carrefour S.A.	France	90 803	93 708	Hypermarket/ supercenter	37	1.8%	2.2%	1.9%
14	▼ -2	Lowe's Companies, Inc.	United States	86 377	86 377	Home improvement	1	3.9%	-11.0%	8.9%
15	▲ 1	Tesco PLC	United Kingdom	85 218	86 243	Hypermarket/ supercenter	5	1.4%	4.2%	1.7%
16	▲ 1	Albertsons Companies, Inc.	United States	79 238	79 238	Supermarket	1	5.5%	2.0%	1.6%
17	▲ 1	Edeka-Verbund	Germany	75 930 e**	78 027 **	Supermarket	1	5.5%	6.5%	n/a
29	● 0	The IKEA Group (INGKA Holding B.V.)	Netherlands	42 960	48 180	Other specialty	31	2.6%	0.0%	3.4%
30	▼ -2	Woolworths Limited	Australia	42 005 **	45 303 **	Supermarket	2	1.6%	4.8%	0.2%

Big Issues in Retail

2026 Retail Industry Global Outlook

1

Value-seeking consumers

Value is the new baseline. Retailers surveyed are retooling pricing, private labels, and loyalty to deliver everyday affordability, turning “value” from a price point into a deeper, data-driven relationship with consumers.

2

AI in commerce

AI is moving from pilot to powerhouse. Retailers surveyed are rapidly deploying agentic AI across pricing, personalization, and planning, turning automation into advantage and transforming every touchpoint of the retail experience.

3

Marketing and customer experience

Marketing is going in-house and AI-first. Retailers surveyed are harnessing data, modernizing tech, and scaling retail media to boost profitability, personalization, and customer experience across every channel.

4

Supply chain transformation

Operational agility takes center stage. Retailers surveyed are cutting tech debt, strengthening supply chains, and scaling AI to create smarter, faster, and more resilient operations for 2026 and beyond.

5

Margin management and cost discipline

Retailers surveyed are pursuing profitable growth with discipline, balancing pricing agility, cost control, and consumer trust to strengthen margins and resilience in an environment of rising costs and shifting demand.

Source: Deloitte analysis.

Operators

- There are four basic mathematical operations that can be performed on numbers:

(eg in Excel)

Multiplication: represented by $\times$ or $*$ or $()$

Division: represented by either $\div$ or $/$ or $\frac{x}{y}$ (numerator)
(denominator)

Addition: represented by $+$

Subtraction: represented by $-$

Order: BODMAS

$$5 + 4 \times 9 = 81 \text{ or } 41?$$

continued

Order: BODMAS

$$5 + 4 \times 9 = 81 \text{ or } 41?$$

Division and Multiplication

BEFORE Addition and Subtraction

Brackets (*parentheses*) take precedence over *all* four basic operations. Hence $5 + 4 \times 9$ can be shown as

$5 + (4 \times 9) = 41$ to remove any ambiguity (but brackets are redundant).

– What if we meant to add first? $(5 + 4) \times 9 = 81$

$$(4 + 9) \times 3 =$$

$$\begin{aligned}(4 + 9) \times 3 &= 13 \times 3 \\ &= 39\end{aligned}$$

B	RACKETS	() [] { }
O	RDER POWER OF	$\sqrt{\quad}$ $(\quad)^2$
D	IVIDE	/ ÷
M	ULTIPLY	* X
A	DDITION	+
S	UBTRACTION	—

Multiplication

- There are several ways to indicate that two numbers are to be multiplied

$$4 \times 6 \text{ or } 6 \times 4$$

$$4 * 6$$

$$(4)(6)$$

$$4(6) \text{ or } (4)6$$

- Multiplying the same signs gives a *positive result*

$$5 \times 6 = 30$$

$$-5 \times -6 = +30$$

$$(-5)(-6) = 30$$

- Multiplying different signs gives a *negative result*

$$(-5)(4) = -20$$

Division

- There are also several ways to indicate that two numbers are to be divided

$$6 \div 3, \quad 6/3, \quad \frac{6}{3}$$

- The number to be divided (6) is the numerator
- The number that is divided by (3) is the denominator

- Dividing the same signs gives a positive result

$$\frac{-6}{-3} = +2$$

- Dividing different signs gives a negative result

$$\frac{-3}{6} = -\frac{1}{2}$$

continued



Order: A & M, S & D

- Addition has symmetry: doesn't matter in which order you add

$$5 + 6 = 6 + 5$$

- If signs identical: use the sign and add

$$-5 + -6 = -(5 + 6) = -11$$

- Subtraction does not have symmetry: order matters!

$$5 - 6 \neq 6 - 5$$

- Multiplication has symmetry: doesn't matter in which order you multiply

$$5 \times 6 = 6 \times 5$$

- Division does not have symmetry: order matters!

(also known as fractions)

$$\frac{6}{3} \neq \frac{3}{6}$$

Calculating Discounts

- A discount off a Recommended Retail Price can be calculated two ways: (eg for 20% discount)

Long way:

Discounted RRP = (old RRP – discount)
Disc. RRP = Old RRP – (old RRP × discount rate)
e.g: $\$80 \times 0.2 = \16 ;
discounted RRP = $80 - 16 = 64$
or: $80 - (80 \times 0.2)$

Quick way:

Discounted RRP = old RRP × (1 – discount rate)
e.g: $\$80 \times (1 - 0.2)$
= $80 (0.8)$
= 64

GST in Australia

[https://www.ato.gov.au/business/gst/when-to-charge-gst-\(and-when-not-to\)/gst-free-sales/](https://www.ato.gov.au/business/gst/when-to-charge-gst-(and-when-not-to)/gst-free-sales/)

10%, except for:

- most basic foods (eg bagels; but bagel chips are taxable; cheese is GST-exempt, and so are unfilled cheese-topped rolls – but filled rolls and cheese platters are taxable)
- some education courses, course materials and field trips
- some medical, health & care services; some medical aids & appliances
- some medicines; some childcare services
- some religious services and charitable activities
- supplies of accommodation and meals to residents of retirement villages
- cars for disabled people to use, as long as certain requirements are met
- water, sewerage and drainage; international transport and related matters
- precious metals; sales through duty-free shops
- grants of land by government; farmland
- international mail; exports
- sales of businesses as going concerns
- some telecommunications supplies
- eligible emissions units.

GST in NZ

<http://www.ird.govt.nz/gst/additional-calcs/calc-spec-supplies/calc-exempt/calc-exempt.html#fs>

15%, except for:

Exempt items (not returned to IRD):

- donated goods sold by not-for-profits
- buying & renting houses;
- financial transactions (eg mortgages, interest; exchanging currency; provision of credit, bank fees)

Zero-rated supplies (returned but not taxed) :

- exports;
 - duty-free goods;
 - sales of going concerns
- Ex-GST: (or $\div 1.15 \times 0.15$)

$$\text{GST} = \frac{\text{GST inclusive price} \times 3}{23}$$

Good example: <http://www.kiwitax.co.nz/infobase/165/how-to-correctly-work-out-the-gst-content-from-a-gst-inclusive-figure>

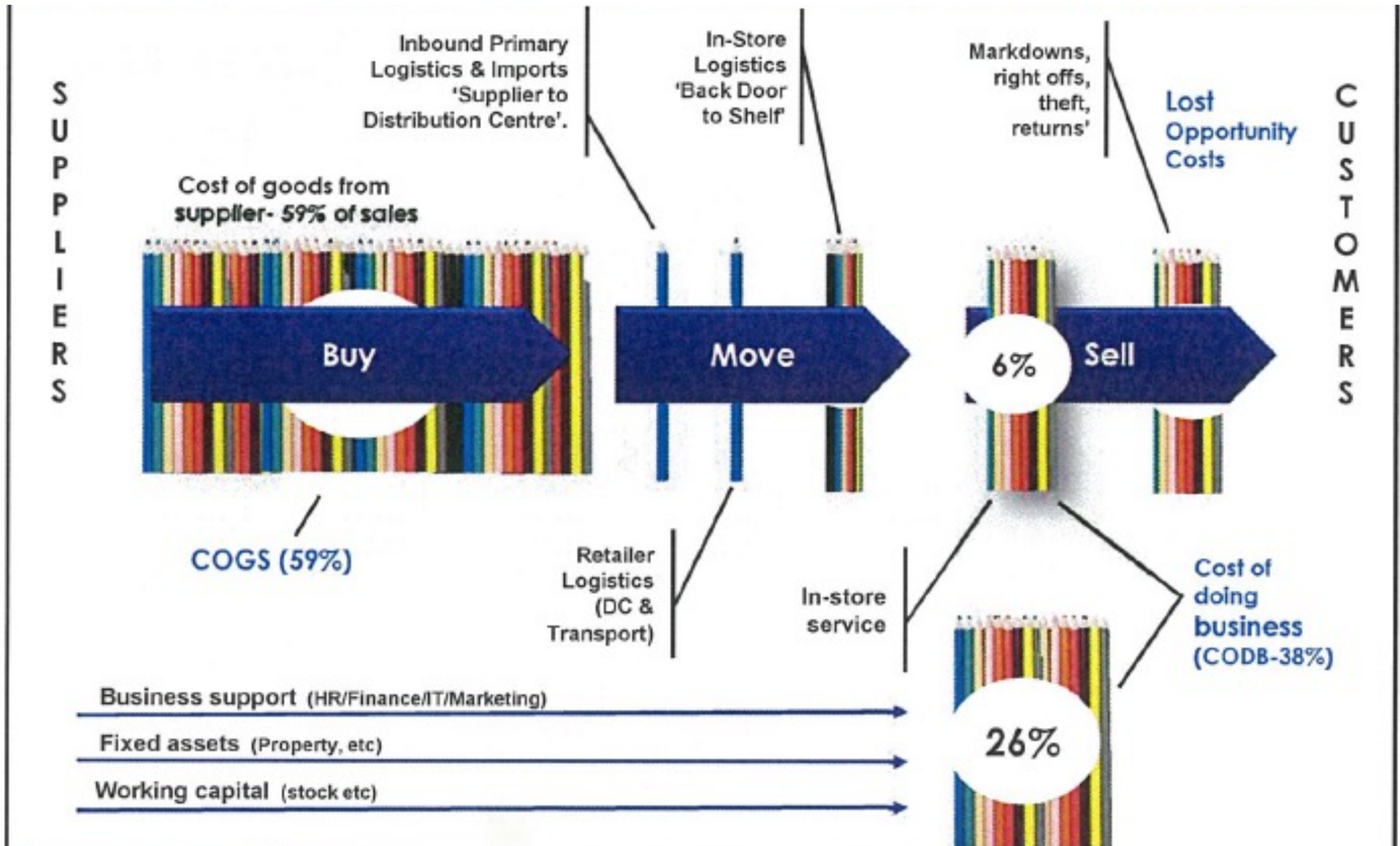
Retail Value Chain

How is value created?

Providing goods and services demanded by customers in a place, time and manner they will pay for, that exceeds the cost of provision. Comprises:

- ‘Buy’ costs: “Cost of Goods Sold” (COGS); shipping costs to NZ; **landed costs** (import duties etc); **terms of trade discounts**
- ‘Move’ costs: DC storage and transport; in-store costs to shelf; **promotions, scanbacks, rebates**
- ‘Sell’ costs: “Cost of Doing Business”: in-store labour; fixed assets (store properties and fixtures); inventory; services provided in-store; losses (from markdowns, theft, returns...) **“burn”**

Retail Value Chain



Calculating GST

- Price excluding GST:

$$\text{ex-GST} = \frac{\text{Sale price aka RRP}}{1.15}$$

- GST to add:

$$\text{GST} = \text{ex-GST price} \times 0.15$$

- GST from RRP:

$$\text{GST} = \frac{\text{RRP} \times 0.15}{1.15}$$



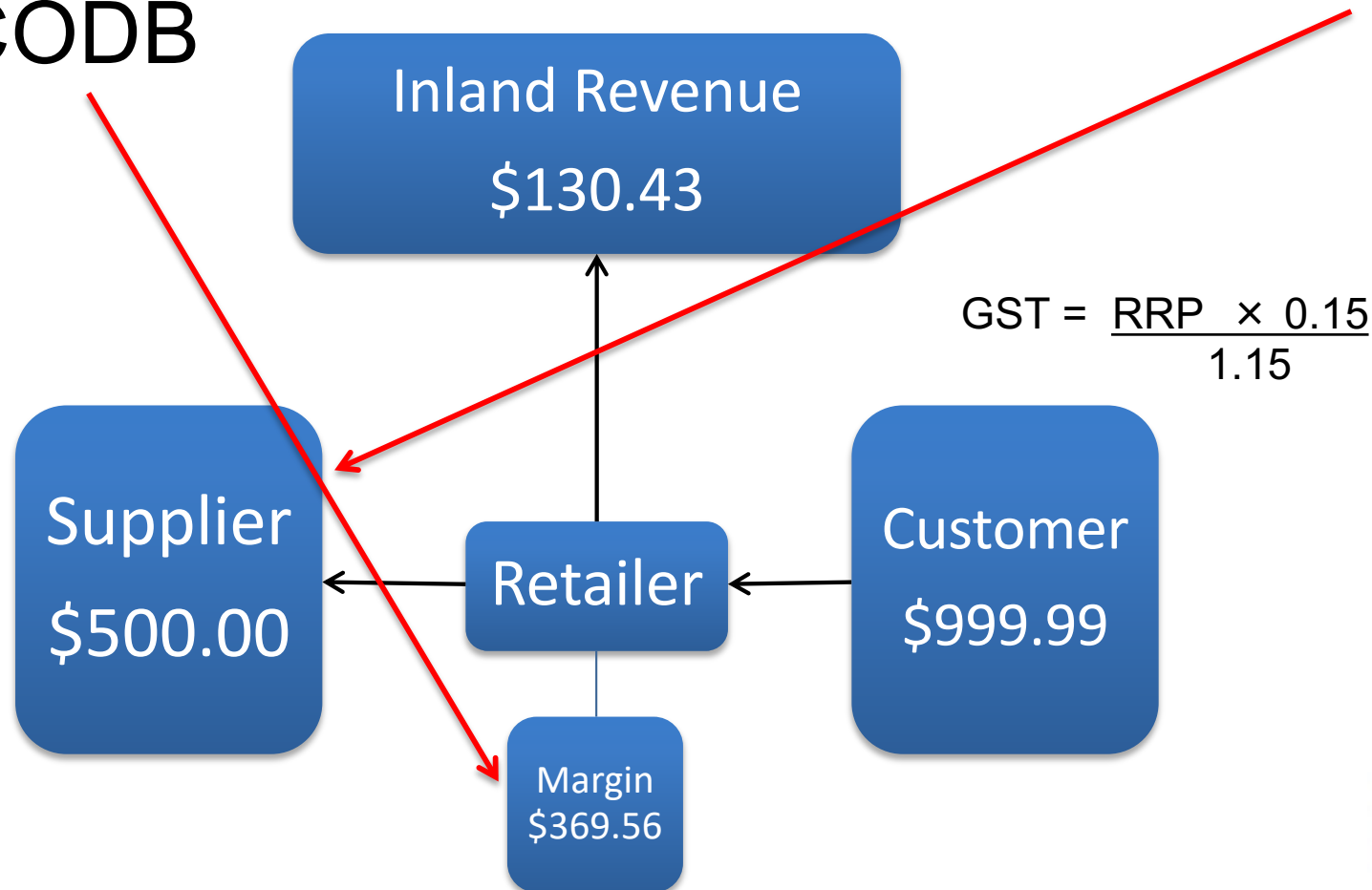
B4		x	✓	<i>fx</i>	=B3*0.15
	A	B	C		
1			rounded		
2	RRP	999.99			
3	Price ex-GST	869.5565217	869.56		
4	GST	130.4334783	130.43		

$$\text{Price ex GST: } \frac{\text{RRP}}{1.15} = \$869.56$$

$$\text{GST} = \frac{\text{RRP} \times 0.15}{1.15} = \$130.43$$

Why report ex-GST?

Part of sale price that belongs to the government: not available to pay for COGS or CODB



Gross Margin (GM)

Contribution made by selling a good or service: \$ made

- Ex-GST Sale (net) price – COGS
- Helps pay the bills: labour, fixtures, property, support functions...

GM %: $GM \div \text{sales} \times 100$

Eg: sell for \$2 (ex-GST); COGS \$1

Gross Margin = \$1

$$GM\% = \frac{1}{2} = 50\%$$



Markup

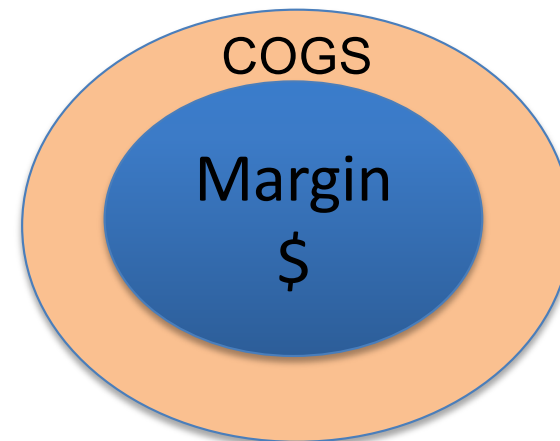
Contribution expressed relative to COGS
= cost-plus

Markup%: $GM \div COGS \times 100$

Eg: sell for \$2 (ex-GST); COGS \$1

Gross Margin = \$1

Markup% = $\frac{1}{1} = 100\%$



Markup vs Gross (Profit) Margin

Good example: <http://www.hardwarejournal.co.nz/feature-stories/march-2012/money-matters-mark-up-versus-gross-profit-margin-there-is-a-difference>

MARGIN % REQUIRED	MARK UP % TO BE PUT ON
10	11.1
12	13.6
15	17.7
20	25
25	33.3
33	50
50	100

Landed Costs

NZ\$ cost of goods once landed in NZ:

- COGS (in origin currency eg RMB) $\div$ exchange rate
- Shipping costs (usually USD) $\div$ exchange rate
- Importation costs (import entry transaction fees, biosecurity system entry fee, customs duties...)
- Inland costs eg transport to DC

Allows a variety of international & domestic suppliers to be directly compared



Terms of Trade (functional) Discounts

Reductions in cost price of goods offered by supplier:

- to increase volume of order or clear stock
- to build their market share
- reflects negotiating power of retailer

Applied prior to invoicing (ie not settled monthly post-hoc)

Weighted Average Cost (WAC)

Shipment costs
vary over time:

- COGS varies with order size
- Shipping costs vary with demand
- Exchange rates vary widely

NZD to USD Chart

New Zealand Dollar to US Dollar

1 NZD = 0.615084 USD Jul 18, 2022, 23:04 UTC



Order	Units Ordered	Cost per unit \$	Total Cost of order
1	50	240.00	\$12,000.00
2	100	200.00	\$20,000.00
3	100	220.00	\$22,000.00
4	50	280.00	\$14,000.00
Total	300	226.67	\$68,000.00

Scanbacks, Rebates & Cash Profit

Reductions in effective cost of goods offered by (“extracted from”) supplier after order:

- Scanback: based on volume sold
Cost price x supplier discount = effective cost price
- Rebate: based on volume ordered (units or total spend)

- Cash profit:

Sales \$ (ex-GST) – Cost of Goods Sold
+ Scanbacks + Rebates = Cash Gross Profit \$

÷ Sales \$ x 100 = Cash Gross Profit %

Burn



Reductions in effective contribution due to promotions, markdowns & clearance

- Poor seller (replaced by another line)
- Seconds (returns or damage by customers / staff)
- End of season (few further sales – inventory carrying costs remain)
- End of life (new range coming in)
- \$ = Original GP\$ - new GP\$
- % = original GP% - new GP%

Effect on Sales & Profit Requirements



Increase in sales \$ & unit sales to maintain original GP\$

		Discount off original price																							
		-2.5%		-5.0%		-10.0%		-15.0%		-20.0%		-25.0%		-30.0%		-35.0%		-40.0%		-45.0%		-50.0%			
		\$	Units	\$	Units	\$	Units	\$	Units	\$	Units	\$	Units	\$	Units	\$	Units	\$	Units	\$	Units	\$	Units		
Current GP%	5.00%	95%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
	10.00%	30%	33%	90%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
	15.00%	17%	20%	43%	50%	170%	200%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
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	55.00%	2%	5%	5%	10%	10%	22%	17%	38%	26%	57%	38%	83%	54%	120%	79%	175%	120%	267%	203%	450%	450%	1000%	0%	0%
	60.00%	2%	4%	4%	9%	6%	20%	13%	33%	20%	50%	29%	71%	40%	100%	56%	140%	80%	200%	120%	300%	200%	500%	0%	0%
	65.00%	1%	4%	3%	8%	6%	18%	11%	30%	16%	44%	22%	63%	30%	86%	41%	117%	56%	160%	79%	225%	117%	333%	0%	0%
	70.00%	1%	4%	2%	8%	5%	17%	8%	27%	12%	40%	17%	56%	23%	75%	30%	100%	40%	133%	54%	180%	75%	250%	0%	0%
	75.00%	1%	3%	2%	7%	4%	15%	6%	25%	9%	36%	13%	50%	17%	67%	22%	88%	29%	114%	38%	150%	50%	200%	0%	0%
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	95.00%	0%	3%	0%	6%	1%	12%	1%	19%	1%	27%	2%	36%	2%	46%	3%	58%	4%	73%	5%	90%	6%	111%	0%	0%

Eg. Taking 25% off the retail price (at a 50% margin) needs a 50% increase in sales and a 100% increase in units to achieve the original GP\$.

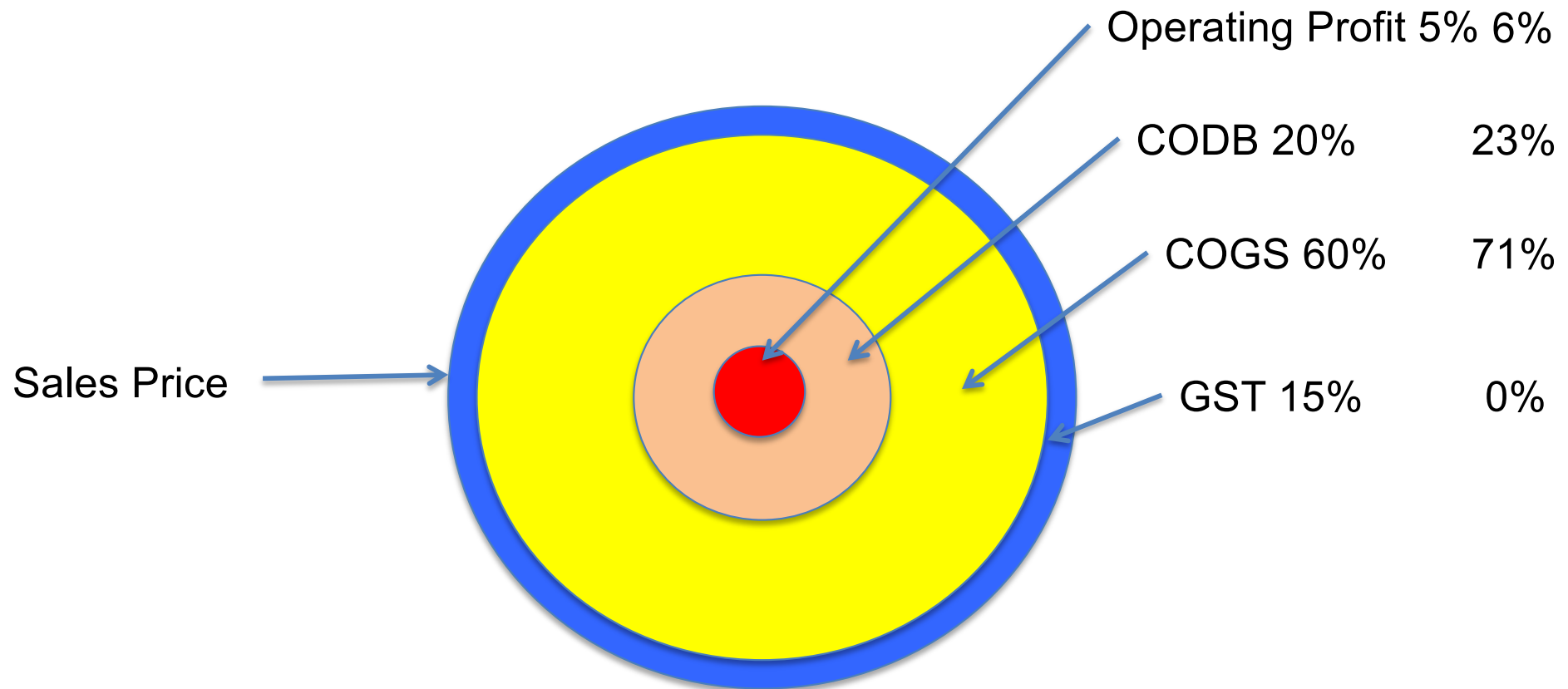
If you were originally planning to do \$100K sales at 50%GP.

after taking 25% off the retail price you will have to do \$150K sales and double the amount of units to achieve the original \$50K GP\$.

Summary:

GM = core business

Ex-GST:



Corporate Level

Consolidated Income Statement

For the 53 week period ended 3 August 2025 (Ex-GST)

	Note	(53 Weeks) 2025 \$ 000	(52 Weeks) 2024 \$ 000
Continuing operations			
Retail sales	2.1	3,086,725	3,037,597
Cost of retail goods sold	8.1	(2,091,643)	(2,016,731)
Gross profit		995,082	1,020,866
Other income	3.1	14,314	7,943
Employee expense	3.2	(526,520)	(512,146)
Depreciation and amortisation expense	3.3	(156,524)	(158,558)
Other operating expenses	3.4	(285,836)	(290,284)
Operating profit from continuing operations	2.1	40,516	67,821
Unusual items	5.0	-	(8,883)
Earnings before interest and tax from continuing operations		40,516	58,938
Interest on leases	10.2	(36,847)	(36,527)
Other net interest	3.6	(6,668)	(1,850)
Profit/(loss) before tax from continuing operations		(2,999)	20,561
Income tax benefit/(expense)	4.1	572	(14,021)
Net profit/(loss) for the period from continuing operations		(2,427)	6,540
Discontinued operations			
Loss from discontinued operations (net of tax)	17.0	-	(60,304)
Net loss for the period		(2,427)	(53,764)

} CODB 31.4%

Corporate Level

Consolidated Income Statement (Ex-GST)

For the 52 week period ended 28 July 2024

	Note	2024 \$ 000	2023 \$ 000
Continuing operations			
Retail sales	2.1	3,037,597	3,236,912
Cost of retail goods sold	8.1	(2,016,731)	(2,148,681)
Gross profit		1,020,866	1,088,231
Other income	3.1	7,943	8,328
Employee expense	3.2	(512,146)	(535,770)
Depreciation and amortisation expense	3.3	(158,558)	(151,891)
Other operating expense	3.4	(290,284)	(286,615)
Operating profit from continuing operations	2.1	67,821	122,283
Unusual items	5.0	(8,883)	(13,561)
Earnings before interest and tax from continuing operations		58,938	108,722
Interest on leases	3.6	(36,527)	(34,374)
Other net interest		(1,850)	(2,818)
Profit before tax from continuing operations		20,561	71,530
Income tax expense	4.1	(14,021)	(21,468)
Net profit for the period from continuing operations		6,540	50,062
Discontinued operations			
Loss from discontinued operations (net of tax)	17.1	(60,304)	(20,125)
Net profit/(loss) for the period		(53,764)	29,937

} CODB 31.6%

Lab 1

D2  Version: 1 (09.07)

115.109: Topic 1 Lab Worksheet

Due date: August 3 Version: 1 (09.07)

Task 1:

This bike is a "classic seller" for the "classic rider". It last sold for RRP \$439.99



Formula Bar

rationale Lab 1:
 This first lab uses two different analytics problems (plus two more in Lab 2 - see the second tab at the bottom of the sheet).
 First, we look at Gross Margin, and the impact of GST and discounts on margin calculations. These are pretty basic but fundamental concepts that are the core to retail business profitability. It is vital that retail staff know the difference between margin and markup, and understand how discounts affect the quantity sold required to maintain the same profit contribution.
 Second, we consider a sample of consumer attitudinal data from a survey in New Zealand, collected in 2020. This is a survey dataset about sustainability attitudes, intended purchase behaviours, and knowledge of and support for retailer sustainability practices. We will use this dataset in several of the labs, and in the Report. For Lab 1 we will look at a small part of the dataset, both in questions (columns) and respondents (rows).
 We start by doing some basic data checking and manipulation: parsing and concatenation (splitting and joining cells).
 We will also do some summary statistics (frequencies), using the CountIf function in Excel. More advanced students could use PivotTables, but we will use that tool in a future lab.

Hint: don't use a calculator! This spreadsheet is a far more capable calculator than anything you have on your phone etc. Use it! Start with typing = in cell E20

Boxes highlighted in yellow will turn green if you enter the right answer (or close enough, if rounding could affect precision of answer). If it stays yellow, check your answer!

Note: as you type something in a cell, that text will appear in the formula bar, above the column headers

E20  =439.99/1.15

	Answer	Show Formula Used
1.1 What is the ex-GST price? Hint: =439.99/1.15	382.60	
1.2 What is the GST included in the RRP?		
If it costs \$200 (ex-GST) to add the bike to inventory: COGS: 200		
1.3 What is the Gross Margin \$? (GM)		
1.4 What is the GM%?		
Recalculate 1.3-1.4 if the next purchase order raises the per unit acquisition cost (COGS) to \$300 (formulas no longer required)		
1.5 What is the COGS for the latest order?		(Making sure you understand the COGS)
1.6 What is the Gross Margin \$?		
1.7 What is the Gross Margin %?		
1.8 What is the markup %?		

Task 2:

Hint: to show a formula instead of calculating it, put a single quote mark in front of the calculation: eg '=E20*15' (to calculate GST). This converts a formula into text.

Hint: change decimal places so only two show for currency: Home ribbon 

Hint: use ex-GST sale price, less COGS = E20-E23

Hint: GM/ ex-GST sale price * 100 (or use the % format, which does this bit for you). Format to 1 decimal place 

This is a "red box": enter this answer into Red Box Quiz for Lab 1 (ignore %). You know whether it is right or not until after due date for submissions.

Do not use the % formatting, as the box will not turn green if you do, even if it shows the right answer

Hint: GM / COGS * 100

DON'T copy and paste into pre-formatted the formatting. You CAN copy and paste the cell, but this does not automatically references - so adjust manually after pa

Lab 1

Task 2:

A second model is added, a fully-suspended mountain e-bike.
It has an RRP of \$6999.99, but has a 20% 'launch' discount.



Complete all the boxes!

Complete cells E52 to E61 based on the RRP, and G51:G61 based on the 20% discount.

After weak sales, Torpedo7 then decide to further reduce the price to clear ("burn") the item, and price the bike at \$5099.99 (including GST)

Complete cells H52:H61 based on clearance ('burn') price.

		Full price:	With 20% discount	Clearance price:
RRP:		\$6,999.99		
2.1 What is the ex-GST price?				
% discount for clearance				
If it costs \$4600 to add the bike to inventory:	COGS	4600	4600	4600
2.2 What is the Gross Margin \$?				
2.3 What is the GM%?				
Units sold at that RRP:		10	20	30
2.4 What is the total sales revenue? (ex-GST)				
2.5 What is the total gross margin \$?				
2.6 and %?				
2.7 Was the discount worthwhile in profit terms?	Answer:			

Enter the new clearance price, then calculate how much of a discount this is from the original full price RRP.
Hint: $(\text{new price} - \text{old price}) / \text{old price} * 100$ (should be -ve)
Is the discount rate the same whether you calculate the discount on the ex-GST or incl-GST price?

Hint: multiply sale price by quantity sold

Murphy, Andrew:
This is a scenario analysis. What happens if 10 bikes sell at RRP?

Calculate the per-unit gross margin
number of units sold

Hint: yes or no, with brief explanation

Task 3: Using the "SustainabilityData-10%sample" datasheet, do some data checks

This is an ~10% subsample: a selection from a larger dataset, which we will use in future labs (and in the report).

For now, we need to check all data is OK (no missing data), and transform column B

Lab 1

90 Select Sort from the Home or Data ribbons (or the Data menu for Macs), and choose Largest to Smallest for column D

91 Depending on which method you choose, you may see the Sort dialog box, or three options only

92 (choosing Custom List... brings up the dialog box).

93 NB: If you only selected column D, Excel would prompt you to Extend the range - but best to select all the data first.

94 9) Interpret the result: any unusually long or short completion times (in min)? What is the longest completion time? minutes

95

96 **Task 4: Use XLOOKUP to find the respondent with the shortest time**

97 XLOOKUP is a powerful tool for extracting data from arrays (columns or rows of data). It combines the previous HLOOKUP (for rows) and VLOOKUP (for columns) into one, and can return multiple results (ie, an array).

98 1) Which response had the shortest response time? You could look at the sorted data and figure out which row and therefore lookup the response ID, or use the XLOOKUP formula.

99 2) The function uses a 'lookup_value' (in our case, the shortest time), a 'lookup_array' (the data it compares that value to: cells D5:D108), and a 'return_array' (the value we want: the corresponding response ID in cells A5:A108)

100 3) First find the shortest time, and enter this into cell C117 of the SustainabilityData-10%sample sheet. You can enter the value itself from the sorted data, or use the formula `=min(D5:D108)`

101 4) Then in cell D117 type the following formula: `=XLOOKUP(C117,D5:D108,A5:A108)` and copy the result here: (Enter response number only in Red Box Quiz: drop the leading P)

102 (NB: The lookup_value could be a fixed value, rather than the cell reference shown above, but in general it is best to avoid fixed values in formulas.)

103

104 **Task 5: Using the "SustainabilityData-10%sample" datasheet, get frequencies of some demographic data using COUNTIF**

105 1) Determine the gender composition of the survey respondents. In cell E4, we can see there are three options: 1=Male, 2=Female and 3=Gender diverse (eg transgender, intersex)

106 To determine how many males we have, we could sort the whole dataset by column E, and then work out how many 1s there are. Or we can use a function: **Countif**

107 2) In cell E110 of the SustainabilityData-10%sample sheet, enter the formula `=COUNTIF(E5:E108,1)`, which counts how many cells in the range above contain a "1".

108 3) Copy the formula in E110 of the SustainabilityData sheet, and paste in E111 of that sheet. Note what happens: because you have gone down a row, Excel changes the range accordingly (`=COUNTIF(E6:G108,1)`)

109 4) To correct this, in E110 add \$ before the row numbers `=COUNTIF(E$5:E$108,1)` and copy and paste again to E111, and E112

110 The \$ anchors a cell reference: in front of a letter, it fixes the column; in front of a number, it fixes the row. (You can fix both at the same time if you wish: `$E$5`)

111 5) Now change the ,1) in the E111 formula to ,2); and in E112 to ,3), to find how many 2s and 3s there are.

112 6) Copy and paste the results in the table below - but because this table is formatted differently, with rows and columns swapped around, you need to **Paste Special (Transpose)**.

113 [Alternatively, you can simply refer from this sheet to that sheet: I have entered this for you in cell B110 of this sheet - and you can ignore the instructions in the next two lines - just copy Cell B110 and paste into C110 and D110, and update to the app

114 Copy cells E110:E112 from the SustainabilityData-10%sample sheet, and in cell B110 below in **this sheet**, right-click and choose Paste Special > and locate the Paste Special entry (**not** any of the pre-formatted options)

115 In the Paste Special box, select Values and Transpose. Complete the table:

	Males =1	Females =2	Diverse =3	Total
Frequencies	 	 	 	0
% of respondents	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!

119

120 7) Now do the same for columns H, I and J of the SustainabilityData-10%sample sheet. These are 'dummy' variables, where the answer options are either selected or not. In column H this is a 2 (if selected) or blank; in column I a 3 or blank; in column J a 4

121 For now, calculate how many respondents selected these options in cells H5:H108, I5:I108 and J5:J108, into cells H114:J114 in the SustainabilityData sheet. Copy and Paste Special > Values and Transpose those answers below.

122 To calculate the non-selected, you can use the total sample size (104) minus the number selected

123

Question:	Selected	Not selected	Answer coding
Enviro responsible	 	 	2 if selected, otherwise blank
Zero waste	 	 	3 if selected, otherwise blank
Social responsible	 	 	4 if selected, otherwise blank

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